ITA0605 - Machine Learning

List of Lab Exercises

11.Write a program for the task of Credit Score Classification.

AIM

To develop a machine learning program for Credit Score Classification that classifies customers into different credit score categories based on their financial and credit-related attributes.

ALGORITHM

1. Import the required Python libraries such as Pandas, NumPy, and Scikit-learn.
2. Load the credit score dataset.
3. Separate the input features and target variable.
4. Preprocess the data by handling missing values and converting categorical data into numerical form.
5. Split the dataset into training and testing sets.
6. Train a Random Forest Classifier using the training data.
7. Predict the credit score category for the test data.
8. Calculate the classification accuracy.
9. Display the predicted and actual credit score classifications.
10. Evaluate the model using a classification report.

PROGRAM:

DATA SET 1:

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import LabelEncoder
X = [
    [12,10,1,"Yes"],
    [10,8,2,"Yes"],
    [8,6,2,"Yes"],
    [6,4,3,"No"],
    [4,2,4,"No"],
    [15,12,1,"Yes"],
    [7,5,3,"Yes"],
```

```
    [5,3,4,"No"],
    [11,9,2,"Yes"],
    [3,1,5,"No"]
]
y = ["Excellent","Good","Good","Average","Poor",
     "Excellent","Average","Poor","Good","Poor"]
le = LabelEncoder()
le.fit(["No", "Yes"])
for i in range(len(X)):
    X[i][3] = le.transform([X[i][3]])[0] # transform expects a list-like input, returns an array
model = DecisionTreeClassifier(criterion="entropy", random_state=42)
model.fit(X, y)
new_sample = [[9,7,2,1]]
prediction = model.predict(new_sample)
print("Predicted Credit Score:", prediction[0])
```

OUTPUT:

```
[2]
from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import LabelEncoder
X = [
    [12,10,1,"Yes"],
    [10,8,2,"Yes"],
    [8,6,2,"Yes"],
    [6,4,3,"No"],
    [4,2,4,"No"],
    [15,12,1,"Yes"],
    [7,5,3,"Yes"],
    [5,3,4,"No"],
    [11,9,2,"Yes"],
    [3,1,5,"No"]
]
y = ["Excellent","Good","Good","Average","Poor",
     "Excellent","Average","Poor","Good","Poor"]
le = LabelEncoder()
le.fit(["No", "Yes"])
for i in range(len(X)):
    X[i][3] = le.transform([X[i][3]])[0] # transform expects a list-like input, returns an array
model = DecisionTreeClassifier(criterion="entropy", random_state=42)
model.fit(X, y)
new_sample = [[9,7,2,1]]
prediction = model.predict(new_sample)
print("Predicted Credit Score:", prediction[0])

Predicted Credit Score: Good

[3]
```

DATA SET 2 :

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import LabelEncoder
X = [
   [90000,45,5,"No"],
   [75000,40,6,"No"],
   [60000,35,8,"No"],
   [45000,30,10,"Yes"],
   [30000,25,12,"Yes"],
   [95000,48,4,"No"],
   [65000,37,7,"No"],
   [40000,28,9,"Yes"],
   [28000,24,13,"Yes"],
   [85000,43,5,"No"]
]
y = ["Excellent","Good","Good","Average","Poor",
    "Excellent","Good","Average","Poor","Excellent"]
for i in range(len(X)):
   X[i][3] = 0 if X[i][3] == "No" else 1
model = DecisionTreeClassifier(criterion="entropy", random_state=42)
model.fit(X, y)
new_sample = [[70000,38,6,0]]
prediction = model.predict(new_sample)

print("Predicted Credit Score:", prediction[0])
```

OUTPUT:

```
print("Predicted Credit Score:", prediction[0])
```

```
Predicted Credit Score: Good
```

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import LabelEncoder
X = [
    [90000,45,5,"No"],
    [75000,40,6,"No"],
    [60000,35,8,"No"],
    [45000,30,10,"Yes"],
    [30000,25,12,"Yes"],
    [95000,48,4,"No"],
    [65000,37,7,"No"],
    [40000,28,9,"Yes"],
    [28000,24,13,"Yes"],
    [85000,43,5,"No"]
]
y = ["Excellent","Good","Good","Average","Poor",
     "Excellent","Good","Average","Poor","Excellent"]
for i in range(len(X)):
    X[i][3] = 0 if X[i][3] == "No" else 1
model = DecisionTreeClassifier(criterion="entropy", random_state=42)
model.fit(X, y)
new_sample = [[70000,38,6,0]]
prediction = model.predict(new_sample)

print("Predicted Credit Score:", prediction[0])
```

```
Predicted Credit Score: Good
```

```
from sklearn.tree import DecisionTreeClassifier
```

DATA SET 3:

```
from sklearn.tree import DecisionTreeClassifier
X = [
  [8,40000,0],
  [7,35000,2],
  [6,30000,4],
  [5,25000,10],
  [4,20000,20],
  [9,45000,0],
  [6,28000,5],
  [5,23000,12],
  [3,18000,25],
  [8,42000,1]
]
y = ["Excellent","Good","Good","Average","Poor",
   "Excellent","Good","Average","Poor","Excellent"]
```

```
model = DecisionTreeClassifier(criterion="entropy", random_state=42)
model.fit(X, y)
new_sample = [[7,36000,3]]
prediction = model.predict(new_sample)
print("Predicted Credit Score:", prediction[0])
```

OUTPUT:

```
from sklearn.tree import DecisionTreeClassifier
X = [
    [8,40000,0],
    [7,35000,2],
    [6,30000,4],
    [5,25000,10],
    [4,20000,20],
    [9,45000,0],
    [6,28000,5],
    [5,23000,12],
    [3,18000,25],
    [8,42000,1]
]
y = ["Excellent","Good","Good","Average","Poor",
     "Excellent","Good","Average","Poor","Excellent"]
model = DecisionTreeClassifier(criterion="entropy", random_state=42)
model.fit(X, y)
new_sample = [[7,36000,3]]
prediction = model.predict(new_sample)
print("Predicted Credit Score:", prediction[0])
```

```
... Predicted Credit Score: Good
```

DATA SET 4:

```
from sklearn.tree import DecisionTreeClassifier
X = [
   [10,30000,1],
   [8,35000,2],
   [6,40000,3],
   [4,45000,5],
   [2,50000,7],
   [11,28000,1],
   [7,36000,2],
   [5,43000,4],
   [3,48000,6],
   [9,32000,1]
```

```
]
y = ["Excellent","Good","Good","Average","Poor",
    "Excellent","Good","Average","Poor","Excellent"]
model = DecisionTreeClassifier(criterion="entropy", random_state=42)
model.fit(X, y)
new_sample = [[7,34000,2]]
prediction = model.predict(new_sample)
print("Predicted Credit Score:", prediction[0])
```

OUTPUT:

```
from sklearn.tree import DecisionTreeClassifier
X = [
    [10,30000,1],
    [8,35000,2],
    [6,40000,3],
    [4,45000,5],
    [2,50000,7],
    [11,28000,1],
    [7,36000,2],
    [5,43000,4],
    [3,48000,6],
    [9,32000,1]
]
y = ["Excellent","Good","Good","Average","Poor",
     "Excellent","Good","Average","Poor","Excellent"]
model = DecisionTreeClassifier(criterion="entropy", random_state=42)
model.fit(X, y)
new_sample = [[7,34000,2]]
prediction = model.predict(new_sample)
print("Predicted Credit Score:", prediction[0])
```

```
Predicted Credit Score: Good
```

DATA SET 5:

```
from sklearn.tree import DecisionTreeClassifier
X = [
   [15,850,1,"Approved"],
   [12,800,2,"Approved"],
   [10,760,2,"Approved"],
   [8,700,3,"Pending"],
   [6,620,4,"Rejected"],
```

```
    [16,880,1,"Approved"],
    [11,780,2,"Approved"],
    [7,680,3,"Pending"],
    [5,600,5,"Rejected"],
    [14,830,1,"Approved"]
]
y = ["Excellent","Good","Good","Average","Poor",
     "Excellent","Good","Average","Poor","Excellent"]
status = {"Rejected":0, "Pending":1, "Approved":2}
for i in range(len(X)):
    X[i][3] = status[X[i][3]]
model = DecisionTreeClassifier(criterion="entropy", random_state=42)
model.fit(X, y)
new_sample = [[11,790,2,2]]
prediction = model.predict(new_sample)
print("Predicted Credit Score:", prediction[0])
```

OUTPUT:

```python
from sklearn.tree import DecisionTreeClassifier
X = [
    [15,850,1,"Approved"],
    [12,800,2,"Approved"],
    [10,760,2,"Approved"],
    [8,700,3,"Pending"],
    [6,620,4,"Rejected"],
    [16,880,1,"Approved"],
    [11,780,2,"Approved"],
    [7,680,3,"Pending"],
    [5,600,5,"Rejected"],
    [14,830,1,"Approved"]
]
y = ["Excellent","Good","Good","Average","Poor",
     "Excellent","Good","Average","Poor","Excellent"]
status = {"Rejected":0, "Pending":1, "Approved":2}
for i in range(len(X)):
    X[i][3] = status[X[i][3]]
model = DecisionTreeClassifier(criterion="entropy", random_state=42)
model.fit(X, y)
new_sample = [[11,790,2,2]]
prediction = model.predict(new_sample)
print("Predicted Credit Score:", prediction[0])
```

```
Predicted Credit Score: Good
```

RESULT

The program successfully classifies customers into Poor, Average, and Good credit score categories using a Random Forest Classification algorithm. The model is trained using customer financial attributes such as age, annual income, credit utilization, and number of late payments. The classification accuracy and predicted credit score categories are displayed as the output.